7 Write a program to implement Logistic Regression (LR) algorithm in python

Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

data = load_iris()
X_train, X_test, y_train, y_test = train_test_split(
    data.data, data.target, test_size=0.2, random_state=42
)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

print("Predicted Class:", model.predict([X_test[0]])[0])
print("Actual Class:", y_test[0])
print("Accuracy:", accuracy_score(y_test, model.predict(X_test)))
```

Output:

